

### Features

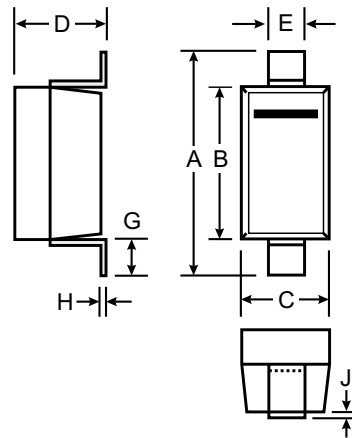
- Fast Switching Speed
- Surface Mount Package Ideally Suited for Automatic Insertion
- For General Purpose Switching Applications
- High Conductance

### Mechanical Data

- Case: SOD-123, Molded Plastic
- Terminals: Solderable per MIL-STD-202, Method 208
- Polarity: Cathode Band
- Marking: Date Code only or Date Code and Type Code

Type Code: T4

- Weight: 0.01 grams (approx.)



| SOD-123              |              |      |
|----------------------|--------------|------|
| Dim                  | Min          | Max  |
| A                    | 3.55         | 3.85 |
| B                    | 2.55         | 2.85 |
| C                    | 1.40         | 1.70 |
| D                    | —            | 1.35 |
| E                    | 0.55 Typical |      |
| G                    | 0.25         | —    |
| H                    | 0.15 Typical |      |
| J                    | —            | 0.10 |
| All Dimensions in mm |              |      |

### Maximum Ratings @ $T_A = 25^\circ\text{C}$ unless otherwise specified

| Characteristic                                                                          | Symbol                          | 1N4148W     | Unit             |
|-----------------------------------------------------------------------------------------|---------------------------------|-------------|------------------|
| Non-Repetitive Peak Reverse Voltage                                                     | $V_{RM}$                        | 100         | V                |
| Peak Repetitive Reverse Voltage<br>Working Peak Reverse Voltage<br>DC Blocking Voltage  | $V_{RRM}$<br>$V_{RWM}$<br>$V_R$ | 75          | V                |
| RMS Reverse Voltage                                                                     | $V_{R(RMS)}$                    | 53          | V                |
| Forward Continuous Current (Note 1)                                                     | $I_{FM}$                        | 300         | mA               |
| Average Rectified Output Current (Note 1)                                               | $I_O$                           | 150         | mA               |
| Non-Repetitive Peak Forward Surge Current @ $t = 1.0\mu\text{s}$<br>@ $t = 1.0\text{s}$ | $I_{FSM}$                       | 2.0<br>1.0  | A                |
| Power Dissipation (Note 1)                                                              | $P_d$                           | 350         | mW               |
| Thermal Resistance Junction to Ambient Air (Note 1)                                     | $R_{\theta JA}$                 | 357         | K/W              |
| Operating and Storage Temperature Range                                                 | $T_J, T_{STG}$                  | -65 to +150 | $^\circ\text{C}$ |

### Electrical Characteristics @ $T_A = 25^\circ\text{C}$ unless otherwise specified

| Characteristic               | Symbol   | Min | Max                           | Unit                                                  | Test Condition                                                                                                                         |
|------------------------------|----------|-----|-------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Maximum Forward Voltage      | $V_{FM}$ | —   | 0.715<br>0.855<br>1.0<br>1.25 | V                                                     | $I_F = 1.0\text{mA}$<br>$I_F = 10\text{mA}$<br>$I_F = 50\text{mA}$<br>$I_F = 150\text{mA}$                                             |
| Maximum Peak Reverse Current | $I_{RM}$ | —   | 2.5<br>50<br>30<br>25         | $\mu\text{A}$<br>$\mu\text{A}$<br>$\mu\text{A}$<br>nA | $V_R = 75\text{V}$<br>$V_R = 75\text{V}, T_J = 150^\circ\text{C}$<br>$V_R = 25\text{V}, T_J = 150^\circ\text{C}$<br>$V_R = 20\text{V}$ |
| Junction Capacitance         | $C_j$    | —   | 2.0                           | pF                                                    | $V_R = 0, f = 1.0\text{MHz}$                                                                                                           |
| Reverse Recovery Time        | $t_{rr}$ | —   | 4.0                           | ns                                                    | $I_F = I_R = 10\text{mA}$ ,<br>$I_{rr} = 0.1 \times I_R, R_L = 100\Omega$                                                              |

Notes: 1. Valid provided that terminals are kept at ambient temperature.

